Antioxidant Activity of Protocatechuic Acid derivatives toward HOO• Radical in Gas phase and Lipid Media: a theoretical study

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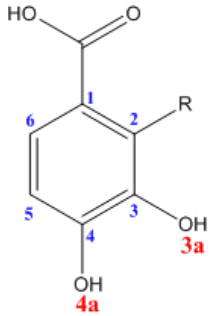


Derivatives

A: R=H (PCA)

B: R=NMe_2

C: R=NHMe

D: R=NH_2

Scheme 1. Molecular structures of PCA

Introduction and Aim

- Antioxidants are chemicals that can provide their own electrons to free radicals to become less reactive in order to reduce the cellular damage [1].
- Lipid oxidation alters the taste and the nutritional quality of the fat also it can lead to toxic compounds like malondialdehyde or 4-hydroxynonenal. To this end, the study of antioxidants that can scavenge free radicals inside the lipid media is of utmost importance [2].
- Protocatechuic acid (PCA) also referred to as 3,4-dihydroxybenzoic acid, is one of the major types of catechol phenolic compounds, widely used in human dies because of its beneficial health effects. Also, it exhibit several pharmacological properties on human health including antibacterial, antiviral, anticancer, anti-inflammatory, antidiabetic, antifungal and antioxidant properties.
- Our aim in this work is to perform a theoretical study of **the antioxidant activity** of PCA and to design new powerful antioxidants derivatives **(B-D)**. Thus, the ***ortho*** position of PCA (compound A, R=H) has been substituted by strong electron-donating groups NMe_2, HNMe, and NH_2 (see Scheme 1) The influence of ***ortho*** substitution on the **thermodynamic and kinetic aspects** of antioxidant activity was also studied in **gas phase** and **toluene** solvent.

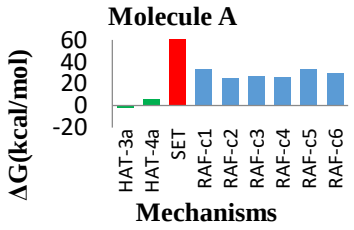
Computational methodology

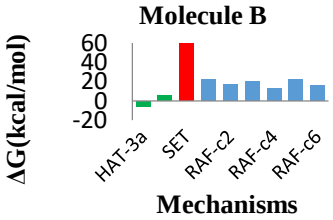
- All computations were carried out using **Gaussian 09** software at the **SMD//M05-2X /6-31+G(d,p)** level of theory.
- The main antioxidant mechanisms considered in this work are **HAT, SET, RAF**.
- The rate constants (*k*) were calculated according to the transition state theory:

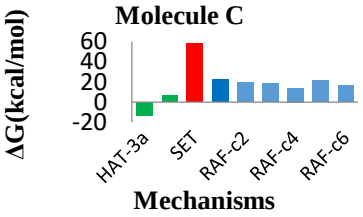
$$k = \sigma\chi(T) + \frac{k_B T}{h} e^{-\Delta G^{\neq}/RT}$$

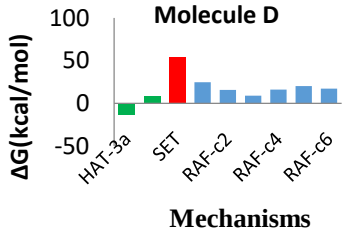
Results and discussion

Thermodynamic study









Kinetic study

Comp	$\Delta G^{\neq}$	K (L.mol-1.s^{-1})
A-3a	15.72	2.05E+02
B-3a	11.21	9.54E+04
C-3a	10.28	7.67E+05
D-3a	09.35	2.80E+06

- **Drug-likeness** evaluation shows that the studied compounds possess satisfactory drug-likeness properties

Conclusion

- The **thermodynamic** results show that the **HAT** mechanism is, thermodynamically the most favoured mechanism in comparison to all the other SET and RAF pathways. And the **3a** position corresponds to the lowest value of ΔG_r for all systems A-D
- The **kinetic calculations** of the exergonic channels show that the designed derivatives B-D react with HOO• radical faster than the reference system PCA (A) and the calculated rate constants in gas phase and in lipid media increase in the following order: **PCA < B < C < D.**

References

[1] Lobo V, Patil A, Phatak A, Chandra N (2010) Free radicals, antioxidants and functional foods: Impact on human health. Phcog Rev, 4:118.

[2] Harles DJ (2012) Natural Antioxidants. In: Antioxidant Properties of Spices, Herbs and Other Sources. Springer New York, New York, NY, pp 39-64.